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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Name of Student:	
Date of Performance	**Date of Completion**

Experiment No. 3

Experiment Name: Introduction to Conditional Statements in JavaScript

Aim/Objective

The objective of this experiment is to **understand and apply JavaScript conditional statements**, including if, if...else, and switch, to create a simple interactive web page that responds to user input.

Pre-requisites

- A text editor (e.g., VS Code).
- A web browser.
- Basic knowledge of HTML and how to include a <script> tag.

Theory/Background

JavaScript is a powerful programming language that makes web pages interactive. **Conditional statements** are a fundamental concept in programming that allow you to execute different blocks of code based on whether a certain condition is true or false.

- **if statement**: This is the most basic conditional statement. The code inside the if block is executed only if the specified condition is true.

```
if (condition) {
  // Code to be executed if the condition is true
}
```

- **if...else statement**: This statement provides an alternative block of code to execute if the condition is false.

```
if (condition) {
  // Code to be executed if the condition is true
} else {
  // Code to be executed if the condition is false
}
```

- **switch statement**: This is a more efficient alternative to a long chain of if...else if statements. It evaluates an expression and executes a block of code based on a matching case.

```
switch (expression) {
  case value1:
```

```
    // Code to execute if expression === value1
    break;
  case value2:
    // Code to execute if expression === value2
    break;
  default:
    // Code to execute if no case matches
}
```

The break statement is crucial to exit the switch block after a match is found. The default case is optional and runs if no other case matches.

Task/Problem Statement

Create a single HTML file named conditionals.html that uses an embedded JavaScript script to perform the following tasks:

1. **if statement**: Check if a user-entered number is positive and display a message.
2. **if...else statement**: Check if a user-entered age is 18 or older and display a message indicating if they are eligible to vote.
3. **switch statement**: Take a number (1-7) representing a day of the week and display the corresponding day name.

Procedure/Steps

1. **Create the HTML file:** Open your text editor and create a new file. Save it as conditionals.html.
2. **Add the HTML structure:** Set up the basic HTML with three buttons, one for each conditional statement demonstration. Use <script> tags to hold your JavaScript code.

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Conditionals</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; margin-top: 50px; }
    button { padding: 10px 20px; font-size: 16px; margin: 10px; cursor: pointer; }
  </style>
</head>
<body>
  <h1>JavaScript Conditional Statements</h1>

  <p>Click the buttons below to run different conditional statement examples.</p>

  <button onclick="checkPositive()">Check if a number is Positive</button>
  <button onclick="checkVoteEligibility()">Check Voting Eligibility</button>
  <button onclick="findDayName()">Find Day of the Week</button>

  <div id="output" style="margin-top: 20px; font-size: 18px;"></div>
```

```
    <script>
        // All JavaScript code goes here
    </script>
</body>
</html>
```

3. **Implement the checkPositive() function:** Inside the <script> tags, write the checkPositive function using an if statement. Use prompt() to get user input.

```
function checkPositive() {
    const number = prompt("Enter a number:");
    const outputDiv = document.getElementById("output");

    if (number > 0) {
        outputDiv.innerHTML = `The number ${number} is a positive number.`;
    } else {
        outputDiv.innerHTML = `The number ${number} is not a positive number.`;
    }
}
```

4. **Implement the checkVoteEligibility() function:** Write the checkVoteEligibility function using an if...else statement.

```
function checkVoteEligibility() {
    const age = prompt("Enter your age:");
    const outputDiv = document.getElementById("output");
    if (age >= 18) {
        outputDiv.innerHTML = "You are eligible to vote!";
    } else {
        outputDiv.innerHTML = "You are not eligible to vote yet.";
    }
}
```

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5. **Implement the findDayName() function:** Write the findDayName function using a switch statement.

```
function findDayName() {
    const dayNumber = parseInt(prompt("Enter a number (1-7):"));
    const outputDiv = document.getElementById("output");
    let dayName;
    switch (dayNumber) {
        case 1:
            dayName = "Sunday";
            break;
        case 2:
            dayName = "Monday";
            break;
        case 3:
            dayName = "Tuesday";
            break;
```

```
        case 4:
            dayName = "Wednesday";
            break;
        case 5:
            dayName = "Thursday";
            break;
        case 6:
            dayName = "Friday";
            break;
        case 7:
            dayName = "Saturday";
            break;
        default:
            dayName = "Invalid number. Please enter a number between 1 and 7.";
    }
    outputDiv.innerHTML = dayName;
}
```

6. **Save and Run:** Save the file and open it in your web browser. Click each button to test the different conditional statements.

Expected Output

When you click each button, a prompt will appear asking for input. After entering the value, a message will be displayed below the buttons, demonstrating the correct conditional logic. For example:

- Clicking the first button and entering 5 should display "The number 5 is a positive number."
- Clicking the second button and entering 20 should display "You are eligible to vote!"
- Clicking the third button and entering 3 should display "Tuesday."

Viva Voce Questions

1. Explain the difference between if...else and switch statements. When would you use one over the other?
2. What is the purpose of the break statement inside a switch block?
3. What does the prompt() function do in JavaScript?
4. How do you ensure that user input from prompt() is treated as a number?
5. What is the default case in a switch statement used for?

Conclusion/Learning Outcome

After completing this experiment, you will be able to write JavaScript code that makes decisions based on conditions. You have successfully implemented if, if...else, and switch statements to create a responsive and dynamic user experience, which is a key skill for building modern web applications.

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